

SQL Questions

Question 1

```
SELECT Employee_Number, Gender, Job_Role, Monthly_Income FROM [HR Data].[dbo].[HR Data] WHERE Department = 'Sales' AND Monthly_Income > 5000;
```

Question 2

```
SELECT COUNT(DISTINCT Job_Role) AS NumberOfJobRole FROM [HR Data].[dbo].[HR Data];
```

Question 3

```
SELECT Department, COUNT(Employee_Number) AS TotalEmployees, AVG(Monthly_Income) AS AverageMonthlyIncome FROM [HR Data].[dbo].[HR Data] GROUP BY Department ORDER BY Department;
```

Question 4

```
SELECT Gender, COUNT(Employee_Number) AS TotalEmployees FROM [HR Data].[dbo].[HR Data] GROUP BY Gender;
```

Question 5

```
SELECT Job_Role, COUNT(Employee_Number) AS TotalEmployees, AVG(Job_Satisfaction) AS AverageJobSatisfaction FROM [HR Data].[dbo].[HR Data] GROUP BY Job_Role HAVING AVG(Job_Satisfaction) < 2.5 AND COUNT(Employee_Number) > 20 ORDER BY AverageJobSatisfaction;
```

Question 6

```
SELECT Department, COUNT(Employee_Number) AS TotalEmployees, SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS ExEmployees, CAST( SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(Employee_Number) AS DECIMAL(10,2) ) AS AttritionRate FROM [HR Data].[dbo].[HR Data] GROUP BY Department ORDER BY Department;
```

Question 7

```
SELECT Employee_Number, Job_Role, Monthly_Income FROM [HR Data].[dbo].[HR Data] WHERE Monthly_Income > ( SELECT AVG(Monthly_Income) FROM [HR Data].[dbo].[HR Data] ) ORDER BY Monthly_Income DESC;
```

HR Data Analysis

Introduction

The purpose of this analysis was to examine employee data and understand the workforce and employee attrition. Power BI was used to explore the data and create visualizations showing different employee characteristics.

The analysis focused on employee numbers, attrition, department, job role, overtime, job satisfaction, monthly income, age and work life balance.

Data Overview

The dataset contained information for **1,470 employees**. The data included employee details such as department, job role, age, monthly income, job satisfaction, overtime, work life balance and attrition.

Analysis and Findings

The analysis showed that **237 employees had left the organization**. This represents approximately **16.12 percent** of the employees in the dataset.

When employee numbers were compared across departments, employees who had left were present in all departments.

When attrition was examined by job role, **Laboratory Technicians had the highest number of employees who left**.

The overtime analysis showed that the number of employees who left was higher among employees who were recorded as **not working overtime**.

For job satisfaction, **level 4 had the highest number of employees** in the dataset. When attrition was compared across satisfaction levels, levels **2 and 4** showed noticeable numbers of employees who left.

The monthly income analysis showed that the largest concentration of employees was around the **2,000 income range**.

The age distribution showed that many employees were around the **early thirties**, with approximately 30 years appearing among the more highly represented ages.

The work life balance analysis showed that **level 3 had the highest number of employees who left**.

Conclusion

The analysis provided a simple overview of the employees and the pattern of employee attrition in the dataset. The main findings were that 237 employees had left, Laboratory Technicians had the highest number of departures, and work life balance level 3 had the highest number of employees who left.

The analysis also showed that most employees were concentrated around the early thirties and around the 2,000 monthly income range.

These findings describe the patterns in the data but do not explain why employees left the organization. Further analysis would be needed to investigate the reasons for employee attrition.